Epidemiologic Study of Prostatic Cancer: Preliminary Report [1]

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SUMMARY

In a preliminary report on a small fraction of the projected subject population, some evidence in support of both sexual activity and venereal transmission hypotheses is accumulating. Prostatic cancer cases, in contrast to hospitalized and/or neighborhood controls, are beginning to show greater proportions of selected sexual activities compatible with venereal transmission of an infectious agent, such as number of sexual partners, use of prostitutes, prior venereal disease, and genital infections in the spouse. Patients with prostatic cancer also appear to have had higher fertility and more prostatic cancer in blood relatives than controls. Age at first intercourse and at first marriage are lower among the cancer patients than among the controls. Antibody titrations for herpesvirus and cytomegalic virus, although currently not revealing striking disparities in positivity, tend to show higher titers among the cancer cases.

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It has been estimated that in 1976 there will be 56,000 new cases of carcinoma of the prostate and 19,300 deaths, thus placing this cancer second among all cancers in men in the United States (1). The Third National Cancer Survey reported a higher incidence of such cancers in the Minneapolis-St. Paul standard metropolitan statistical area than for any of the other areas involved, a rate even higher than for lung cancer, the nation's leading cancer among men (2,3). A number of mortality studies have provided demographic clues for the development of etiologic hypotheses (4–10), but only a few morbidity studies have been done (11–18).[7] Despite the magnitude of the prostatic cancer problem, few analytic epidemiologic studies entertaining specific hypotheses and based on incident cases have been initiated. The relatively higher availability of case material in the Minneapolis-St. Paul area and the paucity of epidemiologic studies in this common cancer site prompted us to entertain a more nearly definitive case-control study.

The epidemiologic evidences, though meager, do point to sexual and venereal activity hypotheses as being worthy of exploration. The sexual activity hypothesis is supported by the higher risk for prostatic cancer among married, widowed, and divorced men and among ever-married men with children (5,9). Patterns of "love-making," included in such an hypothesis, may well explain differences in incidence rates between cultures living in the same environment (19).

Certainly not unrelated to and possibly a component of a sexual activity hypothesis is an hypothesis of venereal transmission. In cervical cancer the incidence among prostitutes and the association of

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[7]Steele R, and Lees REM. Sexual factors in the epidemiology of cancer of the prostate. Presented at the 98th Annual APHA Meeting, 1970.

age at first coitus and number of pregnancies and sexual partners has provided clues for a venereally transmitted virus (20,21), and, recently, cervical cancer patients have been shown more frequently to have antibodies against genital herpesvirus, a venereally transmitted agent with an ability to remain latent within tissues for long periods of time (22).

Whether a parallel exists for prostatic cancer is not known. The possibility of a viral etiology was first intimated by the isolation by Centifanto et al of a herpesvirus from the genitourinary tract of a random population of men without a history of herpesvirus infection (23) and, later, the detection by the same authors of herpes particles in the prostatic cancer tissue of a single patient (24).

The epidemiologic characteristics of the disease, though pointing toward the in-depth exploration of sexual and venereal hypotheses, does not preclude the need for the study of certain other environmental factors and habit patterns such as occupational exposure and dietary items. In this current study these and other factors are being probed, but this strictly preliminary report will, for the most part, deal only with a few selected interesting or provocative variables. It must be clearly understood that none of these factors to be discussed are firm results, because, at best, they represent extremely early findings on but a fifth of the subject population.

METHODS

Case-Control Triads

Two hundred white patients with prostatic cancer from the major hospitals of the Minneapolis-St. Paul area are being collected by a combination of urologist notification and hospital surveillance techniques. The bulk of the case load is being ascertained by two trained nurse-interviewer-abstractors who routinely visit 23 hospitals weekly and have access to admission rosters, surgical schedules, and pathology reports. Permission to approach the patient is sought from the attending physician, the urologist, or both and the patient's permission for interview is sought through a brief description of the study and an approved consent form. Initially, the nurse-interviewers obtain basic demographic data and, with permission, obtain 45 ml of blood by vacutainer for clotting and a total first morning urine specimen. Every effort is made to obtain these specimens prior to surgery if possible or at least between 8 and 10 AM. Specimen collection occasionally must be done in the home and, when necessary, a notation to this effect is made. Following a reasonable post-surgical interval, appointments for the full-scale in-depth interview are made for the male interviewers.

Having ascertained the cancer case, the nurse-interviewers select a hospitalized control in the same hospital, avoiding patients admitted for other genitourinary problems and matching the control by age (± 3 years), race, and similar admission date. The same specimen collection and interview procedures are applied to these controls.

Similarly, to control the socioeconomic factors and because hospital controls may interject selection bias for the factors under study, a second control is selected from the neighborhood of the case by mail, utilizing reverse phone directories, followed by telephone contact for the participation of such a control matched by age (± 5 years) and race. The neighborhood control is sought on the same street of residence as the case, with systematic progression north then south or east then west of the residence of the case until a suitable control is found. The same appointment, specimen collection, and interview procedures are applied as above. Thus, matched triads will be available for analyses.

Specimen Processing

The clotted blood specimens are brought to the laboratory for separation of the sera and partitioning into aliquots for several determinations. The morning urine specimen is measured and two aliquots of 100 ml each are stored in polyethylene bottles; both these and the sera are deep frozen at −80°C for storage. Small quantities of the sera are transferred in batches to the laboratory of Dr. Henry Bauer where herpesvirus and cytomegalic virus antibody determinations are made. Similarly, aliquots will be supplied to the laboratories of the National Center for Disease Control for Epstein-Barr virus antibody titrations. The balance of the sera and urines is stored for future hormonal assay in the hope of determining differences in endocrine patterns between prostatic cancer cases and controls as potential determinants of risk and as screening tools.

Interviews

Male interviewers utilize a structured interview schedule developed with the assistance of Dr. Clyde Martin, formerly of the Kinsey team. An in-depth sensitive approach is utilized by two well-trained interviewers who, in addition to obtaining residential, occupational, and medical and hospitalization histories, probe into smoking and alcohol consumption, use of drugs and medications, marital history,

TABLE 1.—Selected variables among prostate cancer cases, hospitalized controls, and neighborhood controls: preliminary results (Minneapolis-St. Paul standard metropolitan statistical area, 1976)

Variables	Prostatic cancer cases		Hospitalized controls		Neighborhood controls	
	No.	Average	No.	Average	No.	Average
Median age	40	64	43	64	35	66
Residence ever on farm	16/40	40.0%	24/43	55.8%	21/35	60.0%
Prostatic cancer in family	6/36	16.7%	3/41	7.3%	0/35	0.0
History of cirrhosis	1/40	2.5%	4/43	9.3%	0/35	0.0
History of hepatitis	2/40	5.0%	3/42	7.1%	5/35	14.3%
Ever used tobacco	37/40	92.5%	38/43	88.4%	24/35	68.6%
Ever smoked cigarettes	33/40	82.5%	37/43	87.0%	22/35	63.9%
Alcohol use	39/40	97.5%	41/43	95.3%	33/35	94.3%
Median age at first intercourse	39	18.5	42	19.0	35	20.0
Median age at first marriage	37	24.0	42	26.0	35	26.0
Median No. of children	34	3.2	34	2.6	33	2.4
Maximum intercourse in 1 wk	36	4.0	42	5.5	35	7.0
Frequency of intercourse in first yr of marriage (median per wk)	35	3.2	39	4.1	35	4.1
Median No. of sexual partners	40	4.0	42	7.35	35	3.0
Ever used prostitutes	15/40	37.5%	12/42	28.6%	10/35	28.9%
Ever had venereal disease	5/40	12.5%	2/43	4.7%	1/35	2.9%
Genital infections in wife	6/36	16.7%	10/40	25.0%	2/35	5.7%

and detailed patterns of sexual behavior and exposures. These latter questions are designed to distinguish among several patterns of sexual activity if they can discriminate between cases and controls and to provide, along with the designated antibody determinations for candidate viruses, information on possible venereal exposures.

Dietary Questionnaire

A mailed questionnaire on frequency of use of selected dietary items, developed and used successfully in our studies on gastrointestinal cancers, will be applied to all cases and controls.

RESULTS AND DISCUSSION

As of this writing 53 cancer patients, 52 hospitalized controls, and 41 neighborhood controls have had blood and urine specimens collected and 40 cancer patients, 43 hospitalized controls, and 35 neighborhood controls have been interviewed. Because of the small size of the completed sample, matched triad analyses will not be presented in this report. Contrasts or similarities will be based on group characteristics. The number of cases and controls are too few to perform any meaningful cross-tabulations.

Table 1 presents selected characteristics relevant to sexual or venereal hypotheses as well as several other considerations. It should be noted that even without matched-pair analysis the median ages of the case and control groups are virtually identical, indicating reasonably good age-matching procedures. The residential history of the subjects, even with this small size, reveals that fewer prostatic cancer cases than either type of control ever lived on a farm. Although Lilienfeld et al (4) found no urban/rural differences in age-adjusted mortality rates for whites in the United States, King, Diamond, and Lilienfeld had earlier examined the mortality data for metropolitan and non-metropolitan counties by census regions and had found an excess among white residents of metropolitan counties in all but the northeastern region of the country (5). Our data on living subjects thus far are compatible with the latter finding. This would also be compatible with the observation that prostatic cancer occurs less frequently among agricultural workers (5).

Relatively few studies of familial aggregation of cancer of the prostate have been done. Most significant is the one by Woolf among Mormons in Utah in which there was an excess of prostatic cancer mortality among male relatives of prostatic cancer index cases (25). Of immediate interest in our study, despite the small numbers collected thus far, is the finding of an excess of prostatic cancer cases with a history of prostatic cancer in at least one member of the family. Six of the cases (16.7%) revealed cancer of this site as compared to none among 35 neighborhood controls. Of the six cases, four had fathers with prostatic cancer, one had a brother, and one had a grandfather with the disease. One of the four who had a father with this cancer also had two maternal uncles with the disease. Our finding is thus compatible with that of Krain (13) and Steele and Lees[8] who found similar results but also with very small numbers. However, three of the 41 hospitalized controls (7.3%) also indicated that their fathers had had prostatic cancer.

There are an insufficient number of subjects in the study so far to examine a number of other sites of cancer including breast cancer among the female blood relatives. Thiessen (26) found a significantly higher incidence of prostatic cancer among male relatives of breast cancer patients. Wynder et al (27) have indicated a positive correlation between prostatic and breast cancer in a study of international mortality. We are currently collecting data to ascertain whether an excess of breast cancer does exist among the female blood relatives of our subjects as clues to either a common genetic or environmental component for these potentially hormone-dependent cancers.

The hypothesis that cirrhosis of the liver "protects" against prostatic carcinoma is intriguing and is suggested by Robson (28). This hypothesis is compatible with the observed relative mortalities among whites and blacks in the United States in the age groups of greatest mortality from prostatic cancer. The inverse relationship of rates for these two diseases in a country such as Denmark also fits this hypothesis. A biologic mechanism for this relationship may exist in the observation that hepatic cirrhotics, produced either by alcohol or malnutrition or both, have higher than normal estrogen levels (28). In an autopsy study by Glantz a significantly lower proportion of such cirrhotics had evidence of prostatic cancer than did a control group. He suggested the protective effect of hyperestrogenism (29). We are attempting to test this relationship by history of cirrhosis. In table 1 it is obvious that an adequate denominator of subjects does not yet exist. For whatever it may mean, the four cirrhotics among the 43 hospitalized controls are not surprising since this disease is not an exclusion criterion in the selection of matched controls.

Although without apparent relationship between hepatitis (unspecified) and cirrhosis there also are fewer prostatic cancer cases with a history of hepatitis than hospitalized controls and even less than in neighborhood controls. This finding, if maintained to the conclusion of the study, could indicate that altered liver metabolism may be related to prostatic cancer.

With the accumulation of subjects, we shall be able to subdivide for current and discontinued smoking and types and amounts thereof. Of interest at this time is the marked disparity in the proportions designating any use of tobacco between the neighborhood controls (68.6%) and both the cases (92.5%) and hospitalized controls (88.4%). The same general disparities are noted for cigarette smoking. Not only may the similarity of proportions between the cases and the hospital control groups be a reflection of the well-documented greater morbidity rates among smokers than non-smokers (30), but it points up the hazard of utilization of hospital controls for variables which may be related to a number of diseases or to morbidity in general. The merit of a neighborhood control over a hospitalized control is readily apparent here.

Alcohol consumption is also under scrutiny and although "use" data are at this time non-discriminating, frequency and amount of use will be analyzed later.

In structuring an approach to the major hypotheses, namely, types and modes of sexual activity and venereal transmission, a number of variables were considered. Some of these variables could be deemed to be indicators of frequency or intensity and span of sexual activity such as age at first intercourse, age at first marriage, number of marriages, fertility as measured by number of children claimed, maximum times intercourse was performed in any 1 week in the subject's life, number of sexual partners, intercourse with prostitutes, etc. Some of these, such as age at first intercourse, frequency of intercourse, number of sexual partners, use of prostitutes, as well as evidence of venereal disease, genital infections in the wife, types of intercourse or sexual play (such as homosexual encounters), oral or anal sex, and herpesvirus 2 antibody levels, could be indicators of venereal transmission. In addition, the styles of love making, arousal with and without fulfillment, and types and duration of foreplay are also being explored. Time and the small sample size permit only limited analyses of a few of these variables.

In table 1 the median age at which first intercourse was experienced was lower for the prostatic cancer cases (18.5 years) than for either control group (19.0 and 20.0 years). Furthermore, the median age at first marriage was also lower for the cancer cases (24.0 years) than for either control group (26.0 years).

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[8] *See* footnote 7.

TABLE 2.—Distribution of serum antibody titers for herpesvirus and cytomegalic viruses among prostatic cancer cases, hospitalized controls, and neighborhood controls: preliminary results (Minneapolis-St. Paul, 1976)

Titers	Cases		Hospitalized controls		Neighborhood controls	
	No.	%	No.	%	No.	%
Herpesvirus						
≤ 1:8	28	52.8	33	63.5	17	41.5
1:16	5	9.4	9	17.3	10	24.4
1:32	14	26.4	6	11.5	11	26.8
1:64	6	11.3	3	5.8	3	7.3
1:128	0	0.0	1	1.9	0	0.0
Total	53	99.9	52	100.0	41	100.0
Cytomegalic virus						
≤ 1:8	23	43.4	22	42.3	21	51.2
1:16	6	11.3	8	15.4	6	14.6
1:32	10	18.9	13	25.0	10	24.4
1:64	13	24.5	7	13.5	3	7.3
1:128	1	1.9	2	3.8	1	2.4
Total	53	100.0	52	100.0	41	99.9

The ever-married prostatic cancer cases had a greater average number of children (3.2) than either of the two ever-married control groups (2.6 and 2.4). This tends to confirm the findings of Armenian et al with respect to fertility among prostatic cancer cases (15). Also, Greenwald et al reported that fewer prostatic cancer cases than controls had no children (8).

Prostatic cancer cases claimed a smaller median maximum number of times intercourse was performed in any 1 week of their lives (4.0) as compared to either control group (5.5 and 7.0). This finding is compatible with that of Steele and Lees[9] who found that more of the controls than cases reported having more than three coital experiences during the week of maximum activity.

Also, in response to the question "During the first year of your marriage, what was the customary frequency of intercourse?", the median number reported by the cases (3.2) was less than those for either control group (4.1). Both of these questions, according to Martin, correlate well with sexual activity (personal communication). In fact, based on his research, he found that the latter question was the best indicator of "sex drive" or quantity of ejaculatory performance. This is consistent with an hypothesis of stimulation without fruition. In fact from Steele's study[9] information exists that prostatic cancer cases experienced "greater sexual drive estimated in terms of interest in more intercourse than occurred" than did controls.

[9]*See* footnote 7.

A number of variables might indicate venereal transmission; however, some of these may be more significant (eg, visiting prostitutes and a history of venereal disease) than others (eg, number of sexual partners and wives' genital infections). Thus, although the modest excess in the median number of sexual partners (table 1) and the excess of genital infections among the wives of prostatic cancer cases is seen only in relation to the neighborhood controls, a greater proportion of cases (37.5%) than of either control group (28.6% and 28.9%) visited prostitutes and a greater proportion had had venereal diseases (12.5% vs 4.7% and 2.9%).

Studies of the proportions of subjects with antibody against herpesvirus 2, cytomegalic virus, and Epstein-Barr virus as candidates for a viral etiology in prostatic cancer are also under way as indicated above. Preliminary results with herpesvirus and cytomegalic virus are presented in table 2. There appear to be higher proportions of higher titered sera for both herpesvirus and cytomegalic virus antibodies among the cases than among either of the two sets of controls. Quality control has been instituted utilizing blind replicates. We are currently arranging for other herpesvirus 2 antibody tests according to the methods utilized by Aurelian or Rawls or both.

CONCLUSIONS

This is an interim report with several interesting early findings in regard to both sexual activity and venereal transmission hypotheses. Whether the trends noted in a number of the designated variables will assume statistical significance with an increase in sample size remains to be seen. With an increase in the case and control load under study, other variables, particularly with respect to modes of sexual activity, will lend themselves to analysis, and cross-tabulations among confounding and interacting variables will become possible.

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